

Identifying the Drivers of Offensive Success Using ACC TrackMan Data

The goal of this project was to determine which offensive metrics are most strongly associated with season-long offensive success and whether those metrics could be used to create an expected offensive production framework. Traditional statistics such as batting average and OPS tell us what happened, but they do not always explain why it happened. By examining contact quality and plate discipline metrics, this project sought to identify the underlying skills that contribute most to offensive production and create a more complete way to evaluate hitters.

The dataset consisted of more than 250,000 pitch-by-pitch observations collected through TrackMan during the 2026 ACC baseball regular season. Each observation included information on pitch characteristics, pitch location, game context, and batted-ball outcomes. From these observations, player-level metrics were calculated to create season-long offensive profiles.

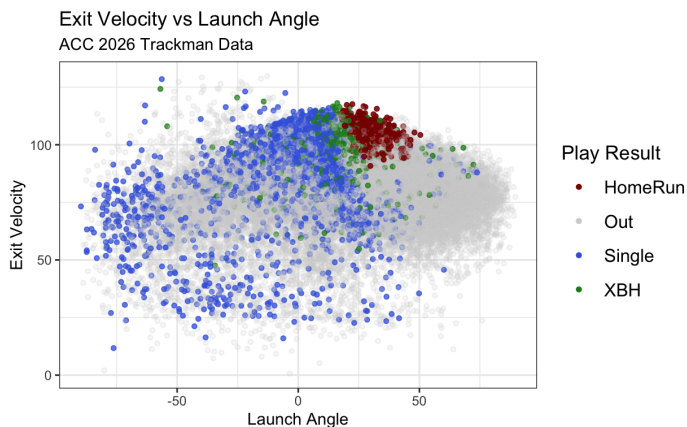
Exploratory analyses showed that the most productive offensive outcomes were generally associated with higher exit velocities and favorable launch angles. Correlation analyses supported these findings, revealing positive relationships between exit velocity and slugging percentage ($r = 0.456$) and launch angle and slugging percentage ($r = 0.370$). Random Forest variable importance analyses further showed that launch angle, exit velocity, and spray direction were the most influential predictors of contact value, while count and pitch location were the most influential predictors of plate discipline outcomes. These findings suggest that both quality of contact and swing decisions play major roles in offensive success.

Random Forest models were then used to estimate contact and plate discipline outcomes. Using a 20% holdout test set, the contact model achieved 76.8% accuracy and the plate discipline model achieved 82.4% accuracy. The outputs of these models were combined to create expected OPS (xOPS), a descriptive metric designed to summarize a hitter's underlying offensive profile. xOPS was highly correlated with actual OPS ($r = 0.931$, $p < 0.001$), suggesting that underlying contact quality and plate discipline metrics explain a substantial portion of offensive performance while also identifying hitters who may have overperformed or underperformed their results.

Several limitations should be considered. The dataset did not include every ACC game because some games were played at venues without TrackMan systems, and observations with missing tags had to be excluded from portions of the analysis. Sacrifice fly opportunities could not be modeled because baserunner situations were unavailable, requiring actual sacrifice fly totals to be used when calculating xOPS. Additionally, defensive positioning data were not available, preventing the estimation of hit probability based on fielder location. Future versions of the framework could incorporate defensive positioning, opponent quality, and ballpark effects to improve estimates of offensive value.

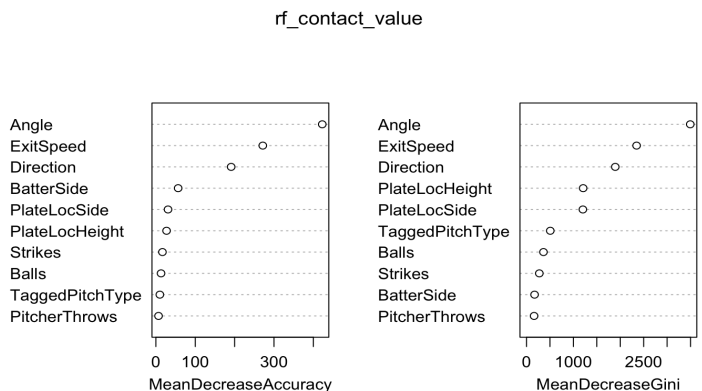
Overall, the results suggest that offensive success is driven by a combination of contact quality and plate discipline rather than any single statistic. Specifically, launch angle, exit velocity, spray direction, count, and pitch location emerged as the variables most strongly associated with offensive outcomes. The xOPS framework provides a useful tool for evaluating underlying offensive performance and identifying hitters whose results may not fully reflect the quality of their process.

EDA 1: Exit Velocity vs Launch Angle



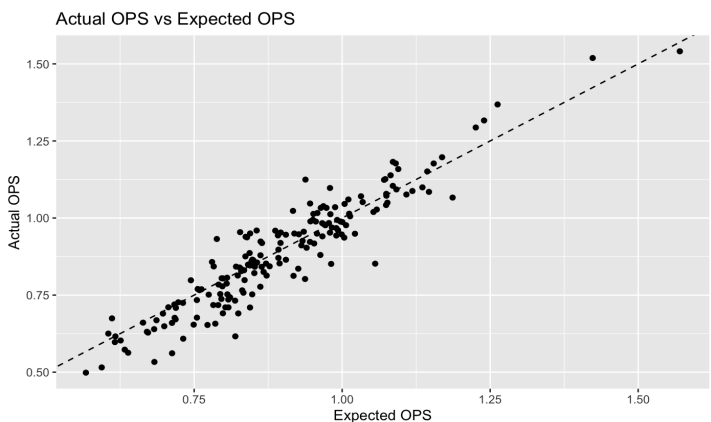
EDA 1: Relationship between exit velocity, launch angle, and offensive outcomes. Extra-base hits and home runs were concentrated in specific combinations of high exit velocity and favorable launch angles, highlighting the importance of contact quality.

RF1: Contact Quality RF Importance Plot



RF1: Variable importance from the contact value Random Forest model. Launch angle, exit velocity, and spray direction emerged as the strongest predictors of contact value.

Actual OPS vs Expected OPS Scatterplot



Relationship between actual OPS and xOPS. The strong positive correlation ($r = 0.931$) indicates that the xOPS framework closely aligns with observed offensive performance while identifying potential over- and under-performers.